

CURRICULUM VITAE

Gifty Attiah

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EDUCATION

- **PhD. Geography** 2020- 2024
Wilfrid Laurier University, Waterloo, Canada
Dissertation: Coupling Remote Sensing and Modelling to Monitor the Spatial Distribution and Trends of Surface Temperature and Ice Thickness on Sub-Arctic Lakes
- **MSc. Environmental Management** 2016 - 2019
Christian Albrecht's - Universität zu Kiel (CAU), Germany.
Project: Vegetation monitoring with Remote Sensing and GIS.
Dissertation: Surface raw material extraction monitoring using Remote Sensing and Machine Learning
- **B.A. Geography and Resource Development** 2010 - 2014
University of Ghana, Ghana
Project: Flood zone detection in Ghana using Remote Sensing
Project: Mapping of hospitals in Accra using GIS for information and accessibility purposes.

PROFESSIONAL APPOINTMENTS

- **Postdoctoral Fellow, Wilfrid Laurier University (WLU)** 10/2024 - present
Overview: Projects on lake water quality monitoring with modelling and Remote sensing
 - Research water quality changes on northern lakes using remote sensing data and machine learning models.
 - Develop geospatial datasets integrating satellite observations and modelling outputs.
 - Supervise undergraduate and graduate students on research projects.
- **Contract Teaching Faculty, Wilfrid Laurier University, Waterloo, Canada.** 01/2023 - present
Courses: Geographical Information Systems (GIS), Computer Applications in Geography, Climate Change
 - Develop and design lectures and facilitated discussions for undergraduate courses.
 - Design practical labs with real-world examples to develop students' technical skills.
 - Provide student support through office hours and academic guidance.

- **Part-Time Teaching Faculty, Conestoga College, Kitchener, Canada** 09/2023 – 04/2025

Courses: Weather and Climate; Understanding Natural Disasters

 - Delivered lectures on weather and climate systems and natural disasters.
 - Designed and graded assignments, exams, and course assessments.
 - Evaluated student progress and performed other organizational duties

- **Graduate Teaching Assistant- Lab Instructor, Wilfrid Laurier University, Canada.** 01/2021- 12/2023

Courses: Remote sensing of Geography, Geographical Information Systems (GIS).

 - Led laboratory sessions and guided students through practical exercises.
 - Assisted students with data analysis, software tools, and lab methodologies.
 - Graded labs and term assessments and conducted other teaching assistant duties.

- **Research Assistant, Remote Sensing of Environmental Change Lab, WLU** 01/2021– 09/2024

Overview: Projects on lake ice thickness monitoring with modelling and Remote sensing

 - Monitoring the changes in lake ice pressure ridges on Great Bear Lake using SAR data.
 - Lake surface temperature monitoring of large Canadian Lakes with MODIS data.
 - Lake ice modelling with a spatially distributed thermodynamic lake ice model.

- **Research Fellow, University of Osnabrück, Remote Sensing and Digital Image Analysis.** 06/2020– 12/2020

Overview: The project involved using remote sensing to monitor long-term agricultural practices

 - Developed a GUI tool's front and back end for satellite image processing and analysis.
 - Conducted fieldwork with Remote Piloted Aircraft (RPAS) for vegetation monitoring.
 - Presented detailed progress reports and performed other research duties.

- **GIS Analyst, Stiftung Naturschutz Schleswig-Holstein, Germany** 08/2018– 03/2020

Overview: Nature Protection Agency working towards the preservation of the environment.

 - Provided GIS and data manipulation support for environmental projects.
 - Developed ArcToolboxes (Python) to automate GIS processes.
 - Generated interactive maps and reports for management and mitigation purposes.

- **Field Supervisor, Centre for Remote Sensing and Geographic Information Services (CERGIS), Ghana.** 03/2016– 06/2016

Overview: The project involved acquiring GIS data on farms to implement sustainable farming.

 - Employed and trained employees to map and measure cocoa farms.
 - Engaged in information and knowledge sharing on sustainable agriculture practices with community members.

- Data Entry and Analyst, *Friends of the Earth- Ghana (NGEO), Ghana*** 12/2015 – 02/2016
Overview: The project was geared towards the establishment of baselines for Volunteer Partner Agreement (VPA) impact monitoring for timber assurance in Ghana
 - Computed and analyzed data collected from field studies across forest areas in Ghana.
 - Developed comprehensive reports based on the analysis of the results and highlighted the impacts of illegal deforestation in communities
- Research and Teaching Assistant, *University of Ghana, Department of Geography & Resource Development*** 09/2014 – 08/2015
Courses: Techniques of Regional Science, Medical Geography
 - Organized and Conducted Tutorial sessions for undergraduate students.
 - Assisted in all stages of conducting various departmental research.
 - Assisted in the development and editing of the Study Guide, slides and grading

TEACHING EXPERIENCE

- | | | | |
|---|---|----------------------------|--|
| • GG368 Computer Applications in Geography | Department of Geography and Environmental Studies | Wilfrid Laurier University | Course Instructor (Winter 2026) |
| • GG389: Climate Change | Department of Geography and Environmental Studies | Wilfrid Laurier University | Course Instructor (Fall 2025) |
| | | | Guest Lecturer (Fall 2024; Winter 2022) |
| • GG369: Geographical Information Systems (GIS) | Department of Geography and Environmental Studies | Wilfrid Laurier University | Course Instructor (Winter 2024; Winter 2023) |
| | | | TA-Lab Instructor (Winter 2022) |
| • GG355: Remote sensing of Geography | Department of Geography and Environmental Studies | Wilfrid Laurier University | Guest Lecturer (Fall 2024; Fall 2023; Fall 2022) |
| | | | TA-Lab Instructor (Fall 2023; Fall 2022; Fall 2021; Winter 2021) |

• LIBS1460: Weather and Climate	School of Interdisciplinary Studies	Conestoga College	Course Instructor (Winter 2025; Fall 2024; Winter 2024)
• LIBS1990: Restless Planet: Understanding Natural Disasters	School of Interdisciplinary Studies	Conestoga College	Course Instructor (Fall 2024; Winter 2024)
• LIBS1135: Blood from a Stone: Social and Environmental Consequences of Mining	School of Interdisciplinary Studies	Conestoga College	Course Instructor (Winter 2025; Fall 2024; Winter 2024)
• SOC1140: Canada in the Future	School of Interdisciplinary Studies	Conestoga College	Course Instructor (Spring 2024)
• GEOG473: Medical Geography	Geography and Resource Development	University of Ghana	Teaching Assistant (1st Semester 2014) (2 nd Semester 2015)
• GEOG452: Techniques of Regional Science	Geography and Resource Development	University of Ghana	Teaching Assistant (1st Semester 2014) (2 nd Semester 2015)

PUBLICATIONS

- **Attiah, G.**, Kheyrollah Pour, H., Scott, K. A. (2023). Lake surface temperature dataset in the north Slave Region retrieved from Landsat satellite series - 1984 to 2021, *Earth System Science Data*, 15, 1329–1355. <https://doi.org/10.5194/essd-15-1329-2023> (IF: 11.2)
- **Attiah, G.**, Kheyrollah Pour, H., Scott, K. A. (2023). Four Decades of Lake Surface Temperature in the Northwest Territories, Canada, Using a Lake-specific Satellite-derived Dataset, *Journal of Hydrology: Regional Studies*, 50, 101571. <https://doi.org/10.1016/j.ejrh.2023.101571> (IF: 5.0)
- **Attiah, G.**, Scott, K.A., Kheyrollah Pour, H. (2025). Spatially Distributed Modelling of Lake Ice Trends and Distribution in the North Slave Region, NWT, Canada, *Water Resources Research*, 61, e2025WR040216. <https://doi.org/10.1029/2025WR040216> (IF: 5.0)
- Persaud, B. D., Shahvaran, A. R., **Attiah, G.**, Chasmer, L. E., English, M. C., Kheyrollah Pour, H., & Wolfe, B. B. (2025). Trend Analysis of Surface Shortwave (Solar) Radiation in Continental Northwest Territories, Canada (1980–2023). *Journal of Climate*, 38(19), 5241–5263. <https://doi.org/10.1175/JCLI-D-25-0123.1> (IF:4.0)

- **Attiah, G.**, Scott, A., Yerubandi, R., & Kheyrollah Pour, H. (2026). Dual-Threshold Models for Turbidity Retrieval on Great Slave Lake using MODIS and VIIRS. <https://doi.org/10.22541/essoar.177275325.55267023/v1> (IF: 5.0) [Preprint]
- **Attiah, G.**, Twum, K.O (2025). Assessing Machine Learning Models for Surface Mining Detection Using Sentinel-2 Data, Remote Sensing, remotesensing-3585227 <https://doi.org/10.20944/preprints202503.2261.v1> (IF:4.2) [Preprint]
- Rafat A., **Attiah G.**, Kheyrollah Pour, H. (2026). The future of ice roads within Canada's Arctic Economic and Security Corridor (AESC). Nature Communications Sustainability [In Review]
- Sekoh A. A., **Attiah, G.**, Kheyrollah Pour, H. (2026). Episodic Discharge Forcing Dominates Turbidity Variability in Great Slave Lake: Evidence from 40 Years of Satellite Observations. Remote Sensingremotesensing-4435174 (IF: 4.2)
- Persaud, B. D., Tia, J., **Attiah, G.**, Bredahl, L., Twum, K.O., Shahvaran, A. R., Kheyrollah Pour, H., & Van Cappellen, P., Rezanezhad F. (2026). From Research to Reach: Evaluating Open Access and Data Availability in Canadian Arctic Scholarship. Canadian Science Publishing [In Review]
- Kheyrollah Pour, H., **Attiah, G.**, Thompson, J., Scott, K.A. Duguay. C (2024). Climogrid V1.0: A Spatially Distributed Thermodynamic Lake Ice Model. Journal of Hydrology, HYDROL66568 (In Review)<http://dx.doi.org/10.2139/ssrn.4979507> (IF: 6.3) [Preprint]
- Twum K.O., **Attiah, G.**, Yeleliere, E., Sasu, D., Boamah, E.F. (2026). Planning for Urban Wetlands in Africa's Coastal Cities: The Case of Urban Communities in Ghana. Landscape and Urban Planning Elsevier. LANDUP-D-25-02712 (IF:9.2) [In Review]
- Twum, K.O , Korah, A., Lartey, D., Asibey, M., **Attiah, G.**, Sarfo, A., Boamah, E. F. (2026), Mapping urban heat dynamics in Sub-Saharan Africa: a multi-city analysis of drivers of urban heat island across African cities. African Geographical Review - 268730509 (IF: 1.0) [In Review]

Datasets

- **Attiah, G.**, Scott, K. A., & Khyerollah Pour, H. (2026). Satellite-Retrieved Modelled Daily Lake Surface Turbidity Data for Great Slave Lake, Canada (2002 - 2025). Federated Research Data Repository. <https://doi.org/10.20383/103.01563> [Dataset]
- **Attiah, G.**, Scott, K. A., & Khyerollah Pour, H. (2025). North Slave Lake Ice Thickness and Phenology retrieved from a thermodynamic model, 1984 to 2022. Borealis, V. <https://doi.org/10.5683/SP3/2FZY9M> [Dataset]
- **Attiah, G.**, Kheyrollah Pour, H., Scott, K.A. (2022). North Slave Lake Surface Temperature Retrieved from Landsat Archives from 1984 to 2021. <https://doi.org/https://doi.org/10.5683/SP3/J4GMC2> [Dataset]
- Persaud, B. D., **Attiah, G.**, Kalamandeen, M., Shahvaran, A. R., Radosavljevic, J., Alves, L., Hamer, E., Maharaj, G., Hastings, R. D., Homa K., P., & Van Cappellen, P. (2025). High-Resolution (1 km) Daily

Climate Gridded Dataset for Guyana (1950–2024). Federated Research Data Repository. <https://doi.org/10.20383/103.01185> [Dataset]

- Kheyrollah Pour, H., **Attiah, G.** (2024). Great Bear Lake and Great Slave Lake Ice Thickness from CLIMoGrid Model. <https://doi.org/10.5683/SP3/KRSJGK> [Dataset]
- Kheyrollah Pour, H., **Attiah, G.** (2024). Great Bear Lake and Great Slave Lake Surface Temperature Retrieved from MODIS Aqua Data. <https://doi.org/10.5683/SP3/O8CCP3>, Borealis, V1 [Dataset]
- Kheyrollah Pour, H; **Attiah, G.** (2022), “Great Bear Lake and Great Slave Lake Bathymetry”, <https://doi.org/10.5683/SP3/BZWZQL>, Borealis, V1 [Dataset]

CONFERENCE PRESENTATIONS

International/National

- **Attiah, G.**, Scott, K. A., Kheyrollah Pour, H. (2026). Leveraging Multi-Sensor Remote Sensing and Modelling to Generate Long-Term Environmental Data for Northern Lakes, Canada Polar Data, Waterloo, Ontario, June 8 – 12 [Paper Presentation]
- Nguyen H., Penn O., Slowinski S., **Attiah, G.**, Hettinga, K., Kheyrollah Pour H., Van Cappellen P., Rezanezhad F. (2026) Effects of Temperature Dynamics on Dissolved Organic Carbon Export from Permafrost Soils. World Wetland Day. Waterloo, Ontario, February 2, 2026 [Poster Presentation]
- Nguyen H., Penn O., Slowinski S., **Attiah, G.**, Hettinga, K., Kheyrollah Pour H., Van Cappellen P., Rezanezhad F. (2026). Effects of Temperature Dynamics on Dissolved Organic Carbon Export from Permafrost Soils. 38th Annual Cold Regions Research Conference, Waterloo, Ontario, January 28 – 30 [Poster Presentation]
- **Attiah, G.**, Scott, K. A., Kheyrollah Pour, H. (2026). Satellite-derived Turbidity Assessment on Great Slave Lake using a Dual-threshold Random Forest Approach, 38th Annual Cold Regions Research Conference, Waterloo, Ontario, January 28 – 30 [Paper Presentation]
- Sekoh A. A., **Attiah, G.**, Kheyrollah Pour, H. (2026). Trends and Analysis of Temperature and Turbidity in the Slave River-Great Slave Lake Mixing Zone Using Google Earth Engine, 38th Annual Cold Regions Research Conference, Waterloo, Ontario, January 28 – 30 [Paper Presentation]
- **Attiah, G.**, Scott, K. A., Kheyrollah Pour, H. (2025). Assessing Lake Turbidity in Great Slave Lake from Satellite Observations Using a Dual-Threshold Random Forest Method, ArcticNet’s Annual Scientific Meeting, Calgary, Alberta, December 15 – 18 [Paper Presentation]
- Sekoh A. A., **Attiah, G.**, Kheyrollah Pour, H. (2025). Trends and Analysis of Temperature and Turbidity in the Slave River-Great Slave Lake Mixing Zone Using Google Earth Engine, ArcticNet’s Annual Scientific Meeting, Calgary, Alberta, December 15 – 18 [Paper Presentation]

- **Attiah, G.,** Scott, K. A., Kheyrollah Pour, H. (2025). Spatially Distributed Modelling of Ice Thickness on Small/Medium Sub-Arctic Lakes using Remote Sensing, 37th Annual Cold Regions Research Conference, Waterloo, Ontario, January 29 – 31 [Paper Presentation]
- **Attiah, G.,** Scott, K. A., Kheyrollah Pour, H. (2024). Lake Ice Thickness Simulation Using a Spatially Distributed Model, ArcticNet's Annual Scientific Meeting, Shaw Center, Ontario, December 9 – 12 [Poster Presentation]
- **Attiah, G.,** Scott, K. A., Kheyrollah Pour, H. (2024). Spatially Distributed Modelling of Ice Thickness on Small and Medium Sub-Arctic Lakes, 7th LAKES 2024 Workshop on Parameterization of Lakes in Numerical Weather Prediction and Climate Modelling, Milan, Italy, November 20 – 21[Poster Presentation]
- **Attiah, G.,** Kheyrollah Pour, H., Scott, K. A. (2024). Remote Sensing and Modelling to Monitor Spatial Trends and Distribution of Surface Temperature and Ice Phenology on Sub-arctic Lakes, Eastern Snow Conference, University of Waterloo, Ontario, June 4 – 6 [Paper Presentation]
- **Attiah, G.,** Kheyrollah Pour, H., Scott, K. A. (2023). Spatially Distributed Modelling of Ice Thickness on Small/Medium Sub-Arctic Lake Using Remotely Sensed Data, American Geophysical Union Annual Science Meeting, San Francisco, Ontario, December 11 – 15 [Poster Presentation]
- **Attiah, G.,** Kheyrollah-Pour, H., Scott, K.A. (2023). Spatially distributed modelling of ice thickness and phenology on Northern Lakes, The VI International AMMCS Interdisciplinary Conference, Waterloo, Ontario, Canada, August 14 – 18 [Paper Presentation]
- **Attiah, G.,** Kheyrollah-Pour, H., Scott, K.A. (2023). Coupling Remote Sensing and Modelling to Investigate Spatial and Temporal Sub-Arctic Lake Ice Changes for The Past Four Decades, 44th Canadian Symposium on Remote Sensing, Yellowknife, Canada, June 19 – 22 [Paper Presentation]
- **Attiah, G.,** Kheyrollah Pour, H., Scott, K. A. (2023). Coupling Remote Sensing and Modelling to Monitor Spatial Trends and Distribution of Surface Temperature and Ice Phenology on Sub-Arctic Lakes, Global Water Futures Fourth Annual Open Science Meeting, Saskatchewan, Canada, May 17 – 19. [Poster Presentation]
- **Attiah, G.,** Kheyrollah Pour, H., Scott, K.A. (2023). Coupling Remote Sensing and Modelling to Monitor Ice Thickness and Phenology on Sub-Arctic Lakes, 36th Cold Regions Research Conference, Ontario, November 23 – 24 [Poster Presentation]
- **Attiah, G.,** Kheyrollah Pour, H., Scott, K.A. (2023). Spatially Distributed Modelling of Lake Ice Thickness and Phenology, ArcticNet Conference Viewing Hub, December 4-7 [Paper Presentation]
- **Attiah, G.,** Kheyrollah-Pour, H., Scott, K. A. (2022). Monitoring changes in surface temperature and ice cover of Canadian sub-arctic lakes from in-situ observation, remote sensing, and modelling, Cryosphere, 2022: International Symposium on Ice, Snow and Water in A Warming World, Reykjavik, Iceland, August 21–26 [Poster Presentation]

- **Attiah, G.**, Kheyrollah-Pour, H., Scott, K. A. (2022). Spatial Variability of Lake Ice Thickness and Phenology on Sub-Arctic Lakes in Yellowknife, Northwest Territories, Canadian Meteorological and Oceanographic Society (CMOS) 56th Congress, the Canadian Geophysical Union (CGU) Annual Meeting, and the 78th Eastern Snow Conference (ESC) 2022 Joint Congress (CMOS-CGU-ESC 2022 Joint Congress), Saskatoon, SK, June 1-3 & 6-8 [Paper Presentation]
- Mansingh, A., Kheyrollah Pour, H., **Attiah, G.** (2022). Projection of Lake Ice Thickness and Phenology under Representative Concentration Pathways (RCP) Scenarios: Great Slave and Great Bear Lake, Northwest Territories. Canadian Meteorological and Oceanographic Society (CMOS) 56th Congress, the Canadian Geophysical Union (CGU) Annual Meeting, and the 78th Eastern Snow Conference (ESC) 2022 Joint Congress (CMOS-CGU-ESC 2022 Joint Congress), Saskatoon, SK, June 1-3 & 6-8. [Poster Presentation]
- **Attiah, G.**, Kheyrollah-Pour, H., Scott, K. A. (2022). Spatial trends and distribution of Surface Temperature and Ice thickness on Sub-arctic lakes using remote sensing and modelling, Global Water Futures Fifth Annual Open Science Meeting, University of Saskatchewan, Canada, May 16 -18, 2022 [Poster Presentation]
- **Attiah, G.**, Kheyrollah-Pour, H., Scott, K. A. (2022). Simulation of Ice Thickness and Phenology on Small Northern Lakes using a Spatially Distributed Thermodynamic Lake Ice Model Coupled with Remote Sensing Observations, The 35th Annual Cold Regions Conference 2022, Waterloo, Canada, November 24 -26. [Paper Presentation]
- **Attiah, G.**, Kheyrollah-Pour, H., Scott, K. A. Gunn, G., Duguay, C. (2021). Modelling the Spatial Variability of Ice Thickness and Phenology on Small Lakes in the North Slave Region, Northwest Territories, 42nd Canadian Symposium on Remote Sensing, Waterloo, Canada, June 21-24. [Paper Presentation]
- **Attiah, G.**, Kheyrollah-Pour, H., Scott, K. A. Gunn, G., Duguay, C. (2021). Simulation of Ice Thickness and Phenology on Small Northern Lakes using a Spatially Distributed Thermodynamic Lake Ice Model Coupled with Remote Sensing Observations, Waterloo, Canada, May 17 -19 [Paper Presentation]
- **Attiah, G.**, Kheyrollah-Pour, H., Scott, K. A. (2020). Spatial Variability of Small Lakes in the North Slave Region, Northwest Territories, Cold Regions Research Conference, Waterloo, Canada, November 26-27. [Paper Presentation]

INVITED TALKS

- **Attiah, G.** (2026). Observation of Temperature and Turbidity on Great Bear and Great Slave Lake, Canada Water Agency, Climate Change Workshop, March 4-5, 2026, Yellowknife, Canada
- **Attiah, G.** (2023). Lake Surface Temperature and Ice Thickness in the North Slave Region, NWT, Laurier Délégation, Wilfrid Laurier University, March 3, 2023, Canada
- **Attiah, G.** (2023). Monitoring Surface Temperature and Ice Thickness on Small/Medium Lakes in the North Slave Region, Aurora College/Wilfrid Laurier Winter Series, February 16, 2023, Canada

- **Attiah, G.** (2022). Monitoring Lake Surface Temperature and Ice Thickness in the North Slave Region, NWT, Government of Northwest Territories-Wilfrid Laurier University AGM, October 3 -4, 2022, Canada

AWARDS AND SCHOLARSHIPS

• Polar Knowledge Canada Fellowship	CA\$50,00.00	2026
• Laurier Graduate Outstanding Work Award		2025
• Graduate Doctoral Studentship	CA\$59,750.00	2021 - 2024
• Differential Tuition Award	CA\$51,144.96	2020 - 2024
• Laurier Graduate Scholarship	CA\$34,000.00	2020 - 2024
• American Geographical Union Travel Award	CA\$1300.00	2023
• Global Water Features Travel Award	CA\$1000.00	2023
• American Geographical Union Cryosphere DEI Award	CA\$650.00	2023
• Laurier Institute for Water Science Graduate Student Travel Award	CA\$500.00	2022
• Cold Regions Research Centre Graduate Student Award	CA\$740.00	2022
• The Cryosphere 2022 Symposium Student Travel Award	CA\$780.00	2022
• FGPS Graduate Travel Award	CA\$1000.00	2021, 2022
• Graduate Student Top-Up Funding	CA\$1000.00	2020

GRANTS

Awarded

- Co-Applicant – Advancing an Interdisciplinary Collaborative Network for Arctic Ice Road Infrastructure and Knowledge Integration
Polar Knowledge Canada - \$177,590 2025 - 2027
- Applicant – Scalable Field-Calibrated Sensing Framework to Quantify Climate Warming Impacts on Water Quality in Small Northern Lakes.
Polar Knowledge Canada - \$102,000 2026 - 2027

Applied

- Co-Applicant – Linking Ice, Water Quality, and Fish Habitat at the Great Slave Lake Watershed Scale for Community Resilience.
Environmental Damages Fund - \$260,000 2026 - 2029

SERVICE AND LEADERSHIP

- **Administrator, Cold Regions Research Centre, WLU** 02/2024 – present
Overview: Perform administrative and other duties for the centre
 - Manage general office tasks like managing emails, scheduling meetings
 - Support various events by assisting with coordination and implementation.

- Provide general administrative support to ensure the smooth operation of the CRRC office

- **Conference Programming Committee, *Canadian Science Policy Conference*** 02/2025 – present
Overview: The CRRC represents the interests of cold regions researchers
 - Part of the organization of the yearly conference

- **Programs Committee, *Cold Regions Research Centre (CRRC)*** 09/2024 – present
Overview: The CRRC represents the interests of cold regions researchers
 - Organization of the yearly conference

- **Reviewer** 2024-present
 - Cryosphere (IF:4.4)
 - Remote Sensing (IF: 4.2)
 - Journal of Hydrology: Regional Studies (IF: 5.0)

- **Judge**
 - Poster Judge: World Wetlands Day. 2025 02/2025
 - Poster Judge: 34th Annual Cold Regions Research Conference 11/2022
 - Poster Judge: 33rd Annual Cold Regions Research Conference 11/2021

- **Session Chair, *Eastern Snow Conference (ESC) 2024*** 06/2024
Overview: ESC is a yearly conference to highlight the works in snow and ice
 - Hosted a session at the conference.

- **Early Career Executive, *Cold Regions Research Centre (CRRC)*** 09/2021 – 08/2023
Overview: The ECR represents the interests of early-career researchers in CRRC
 - Organize monthly seminars and educational activities for CRRC students.
 - Organize yearly conferences (CRRC-Days) to support student research

- **Student Committee Organizer, *42nd Canadian Symposium for Remote Sensing (CSRS) 2021*** 03/2021–06/2021
Overview: CSRS is a yearly conference to highlight the works in the field of remote sensing
 - Organized educational and recreational student activities for the conference.

- **Conference Host, *42nd Canadian Symposium for Remote Sensing (CSRS) 2021*** 07/2021
Overview: CSRS is a yearly conference to highlight the works in the field of remote sensing
 - Hosted various sessions at the conference (e.g., Remote Sensing of Lake Ice)

- **Global Ambassador, *CLIMATE 2020 - The Worldwide Online Climate Conference*** 01/2020 – 03/2020
Overview: A global online conference to educate people on climate change and sustainability.
 - Engaged in publicity and organizational activities to publicize the online conference.

- Communicated with writers and reporters to expand the conference's reach
- **Consultant, *Eco-bridge Consulting*,** 10/2018 –
Overview: Consultation on sustainability for small businesses in Kiel (Germany). 04/2019
 - Developed a tool to help small companies visualize and measure their environmental impact.
 - Provided consultation to small businesses to reduce emissions.
- **Vice President, *Ghana Students Association; Christian-Albrecht's Universität Zu Kiel, Germany*** 2017 - 2019
- **Organizing team, *Performing Change: An Interactive Workshop on Consumerism and Sufficiency; Christian Albrecht's Universität Zu Kiel*** 2018
- **Choreography directress, *Hearts in Service Theatre, University of Ghana (2012-2013)*** 2012-2013
- **Organizing Secretary, *Women in League, University of Ghana*** 2012-2013

SUPERVISION AND MENTORSHIP

- | | | |
|--------------------|-----------------------|----------------|
| • Andy Sekoh | Graduate Student | 2024 - present |
| • Lan Yi | UG Research Assistant | 2025 - present |
| • Mya Formann | UG Research Assistant | 2025 |
| • Moulik Budhiraja | UG Research Assistant | 2023 - 2025 |
| • Parth Chauhan | UG Research Assistant | 2023 - 2025 |
| • Ariana Mansingh | UG NSERC USRA | 2021 - 2022 |
| • Christian Balea | UG Research Assistant | 2020 - 2021 |

TECHNICAL COMPETENCIES

- | | |
|------------------------------------|-------------------------------------|
| • Remotes Sensing | • Machine Learning Modelling |
| • Spatial Distributed Modelling | • Change detection and prediction. |
| • Data Analysis and Manipulation | • Geovisualization and Map Creation |
| • ArcGIS toolbox development | • Spatial data extraction |
| • Satellite Image Processing | • Python Module Development |
| • Spatial and Statistical Analysis | • GUI development |

TECHNICAL TOOLS AND SOFTWARE

Remote Sensing

- | | |
|----------|----------------|
| • Python | • ENVI- Harris |
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- Google Earth Engine
- Geomatica
- GDAL
- Panoply

- SNAP-Sentinel Toolbox
- R
- Google Earth Pro
- Computing platforms

GIS

- ArcGIS Pro
- Arcpy (Python)
- QGIS
- GrassGIS
- AutoCAD Map
- FME
- PostgreSQL
- Geoserver

Others

- Microsoft Office
- SPSS
- Visual Studio Code
- MobaXterm
- Linux (Bash)
- Git
- Selenium
- WinSCP

PROFESSIONAL DEVELOPMENT

Teaching Certifications

- University Teaching Certificate
- University Teaching Foundations Certificate

Technical & Research Training

- Small Advanced RPAS (Drone) Certificate
- Google Earth Engine Workshop, CRSS

Safety & Field Training

- Blended Standard First Aid – Wilderness
- St. John Ambulance First Aid Training
- Young Worker Health and Safety Orientation Certificate

Academic & Professional Skills Workshops

- Academic Writing Workshop
- Academic CV Workshop
- Preparing for Research Showcase
- Transferable Skills: Mobilizing Your Skillset for Master's, PhD Students & Postdoctoral Fellows
- Network with Confidence: Networking for Graduate Students and Postdoctoral Fellows
- Workplace Professionalism for Graduate Students and Postdoctoral Fellows

PROFESSIONAL ASSOCIATIONS

- Canadian Remote Sensing Society (CRSS)
- American Geophysical Union (AGU)

- Canadian Meteorological and Oceanographic Society
- Canadian Science Policy Centre
- Cold Region Research Centre, Wilfrid Laurier University (CRRC)
- Laurier Institute for Water Science

MEDIA OUTPUTS

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| • Laurier student researchers tackle climate change in Canada's North: Video | 04/2023 |
| • Research Chat Podcast: Monitoring Lakes from Space | 06/2022 |
| • Laurier Graduate Student Competition | 06/2021 |

OTHER ACTIVITIES

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| <ul style="list-style-type: none"> • <i>SnowEx Hackweek 2021</i>
Fostered an environment of collaboration and sharing of NASA SnowEx datasets across teams, and created projects using programming | 12-16/07
/2021 |
| <ul style="list-style-type: none"> • <i>Laurier Graduate Student Competition, Wilfrid Laurier University</i>
Shot a 3-minute video highlighting key points of research | 06/06
/2021 |
| <ul style="list-style-type: none"> • <i>Sustainable Entrepreneurship Autumn School</i>
Focused on training students for sustainable entrepreneurship related to SDGs | 1-5/11
/2019 |
| <ul style="list-style-type: none"> • <i>Climate-KIC Journey</i>
The five-week program involved entrepreneurship development- product creation, pitch training, and business proposal writing to propel climate change mitigation and adaptation. | 17-5/08
/2018 |
| <ul style="list-style-type: none"> • <i>Energy Efficiency Hack, 2018</i>
Focused on creating and ideating new energy efficiency solutions, concepts and prototypes to combat climate change. Winning Team: Most Technical Innovation | 18-19/
02/2018 |

LANGUAGES

- English (Professional Proficiency)

REFEREES

- Dr. Robert McLeman
Professor
Department of Geography and Environmental Studies
Wilfrid Laurier University
Email: rmcleman@wlu.ca
Phone: +1(548) 889-3866
(Department Chair and CTF supervisor)
- Dr. Homa Kheyrollah Pour
Associate Professor
Department of Geography and Environmental Studies
Wilfrid Laurier University
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